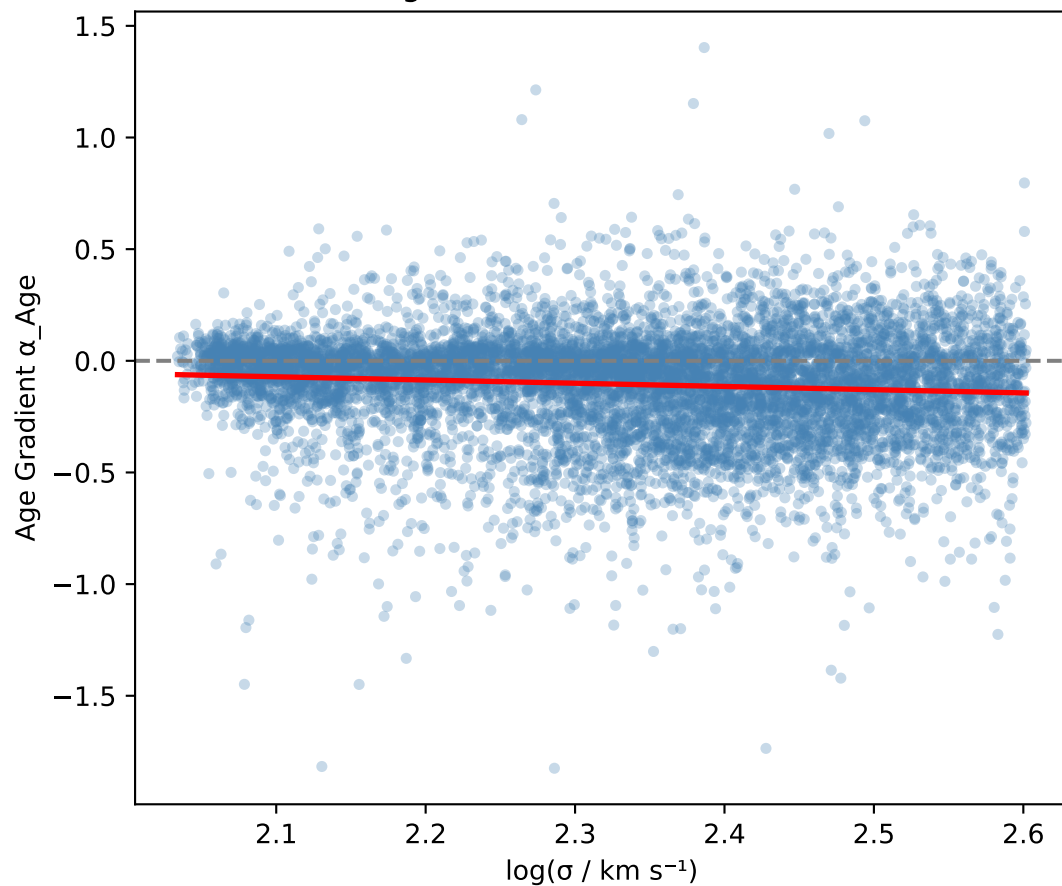
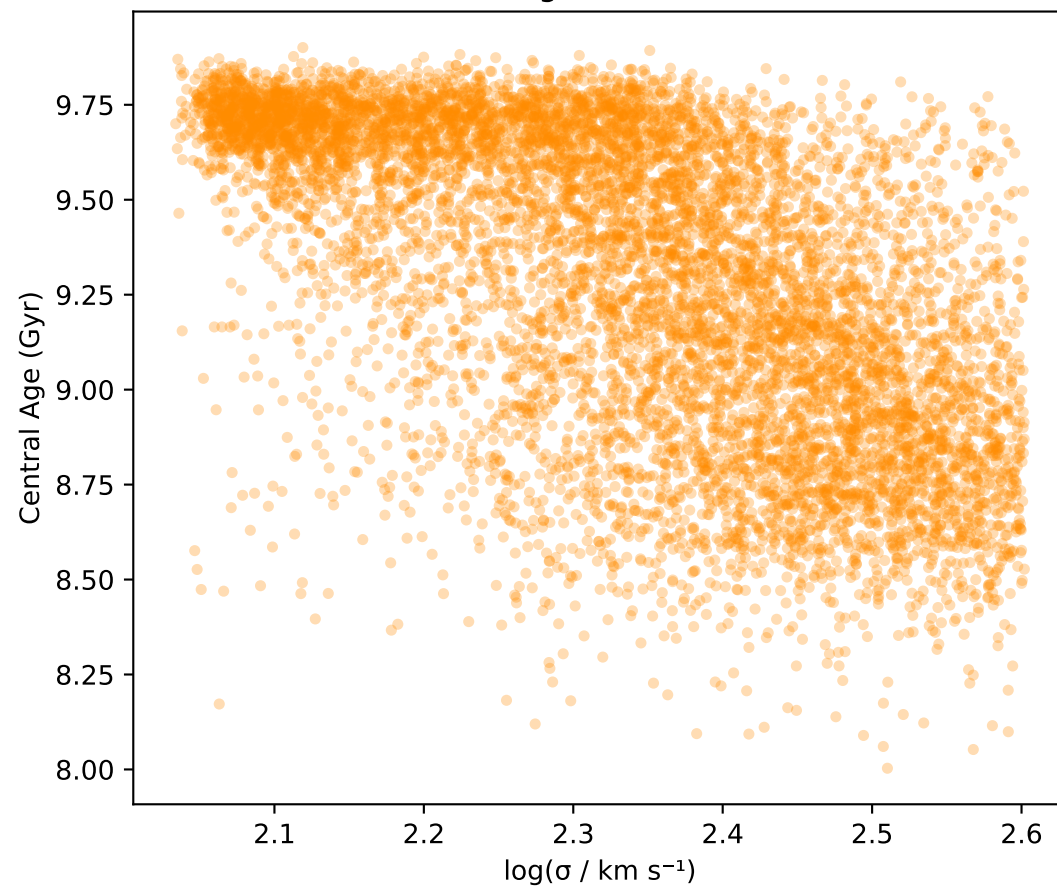
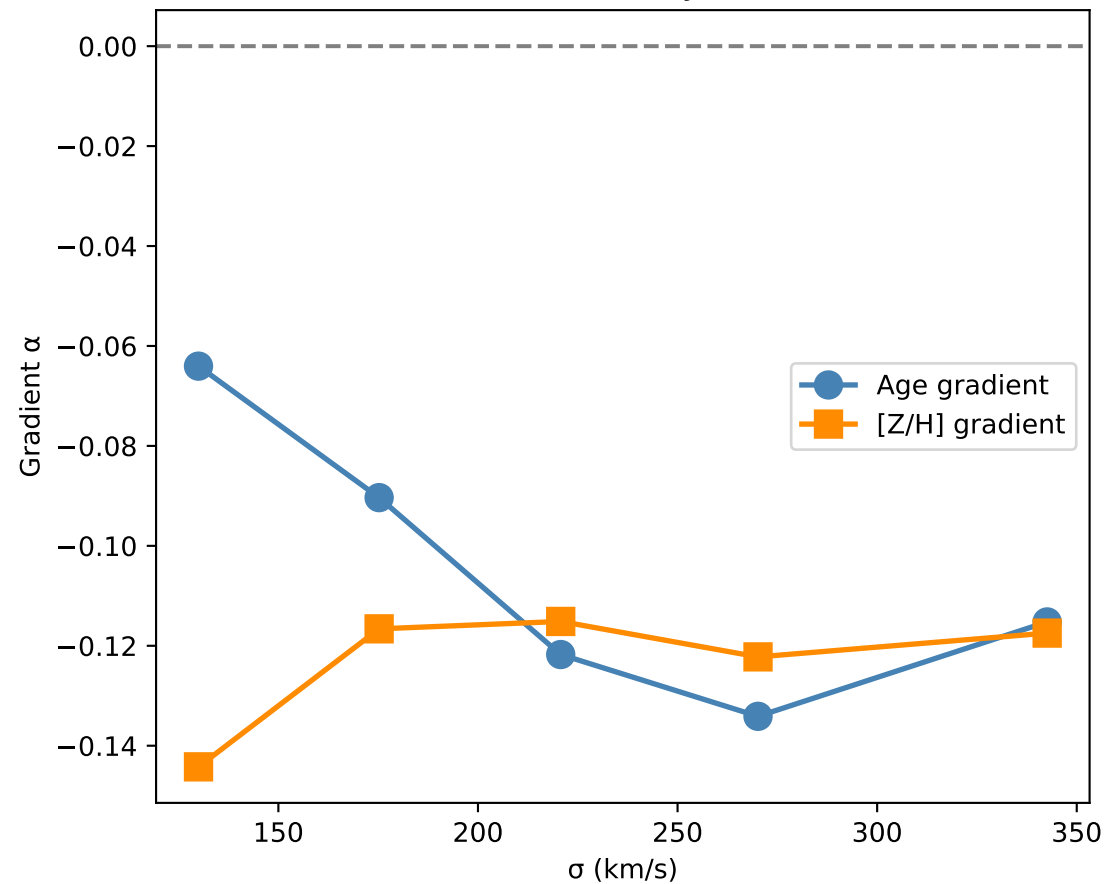


A. Age Gradient vs σ ($r = -0.092$)B. Central Age vs σ ($r = -0.603$)C. Gradients by σ Bin

MaNGA SPATIALLY-RESOLVED TEST

Sample: 8,642 galaxies with IFU data

CORRELATIONS WITH $\log(\sigma)$:

Age gradient: $r = -0.0919$

[Z/H] gradient: $r = 0.0686$

Central age: $r = -0.6032$

INTERPRETATION:

TEP CONSISTENT

TEP PREDICTION:

Negative age gradient (younger centers)
should correlate with higher σ .

$\alpha_{\text{Age}} < 0$ means center is YOUNGER than outskirts.
If TEP causes time dilation in galaxy centers,
we expect α_{Age} to become more negative at higher σ .